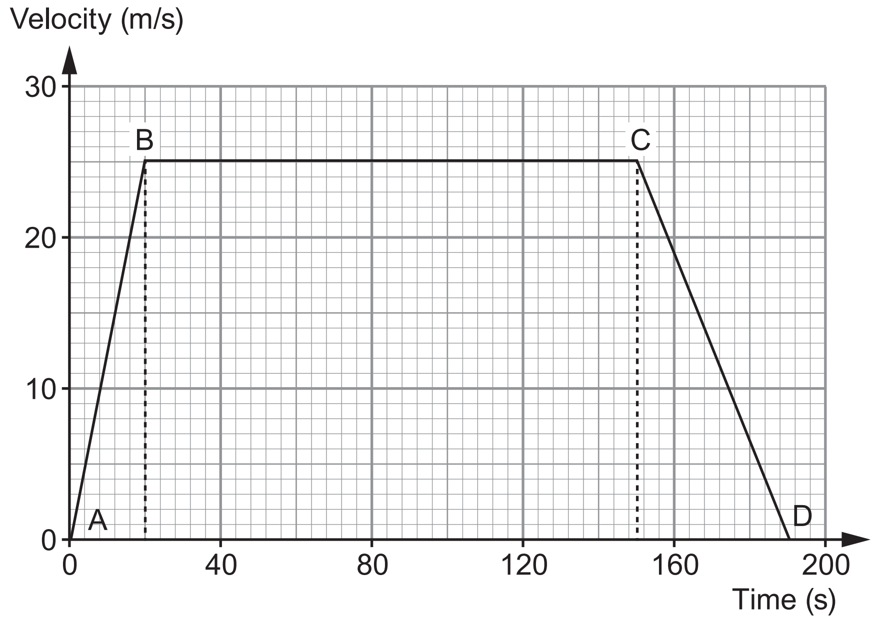
Examiner only

4. An empty school bus takes 190 seconds to travel between two sets of traffic lights. Its journey is shown on the velocity-time graph below.



(a) The bus accelerates uniformly from the first set of traffic lights at A. The resultant force acting on it is 5000 N. Use the graph and equations from page 2 to calculate the mass of the empty bus. [4]

Mass = ... kg

(b) The same resultant force acts on the bus which is now full of pupils. Explain how the start of the velocity-time graph (between A and B) would change. [2]

....

....

....

Examiner only

(c) (i) State Newton's first law of motion. [2]

....

....

(ii) Explain how Newton's first law applies to part **B to C** of the journey. [2]

....

....

....

(d) During its journey from A to D how long is the bus moving at a velocity greater than 12.5 m/s? [2]

Time = ... s

(e) A lorry travels between the same two sets of traffic lights at a constant velocity without stopping. It takes the same time as the school bus. **On the graph opposite** draw the motion of the lorry. You may use equations from page 2 and the space below to show any calculations. [5]

Space for workings:

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